Where Prices Come From: The Interaction of Supply and Demand

ECON 101 – Principles of Macroeconomics

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Spring 2026

Chapter Outline

- **3.1** The Demand Side of the Market
- **3.2** The Supply Side of the Market
- **3.3** Market Equilibrium: Putting Buyers and Sellers Together
- **3.4** The Effect of Demand and Supply Shifts on Equilibrium

3.1 Learning Objective

Goal

Discuss the variables that influence demand.

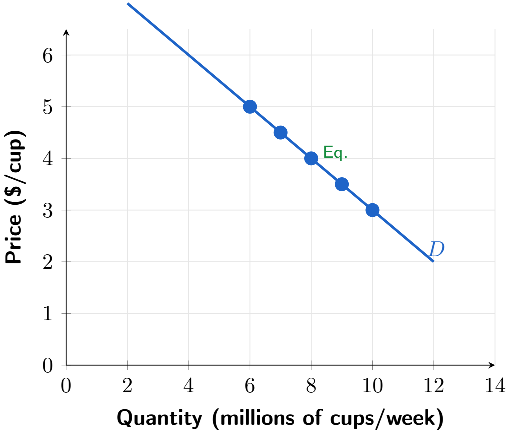
- We begin by investigating how **buyers** behave.
- **Market demand:** the demand by all consumers of a given good or service.
- Key tools: demand schedule and demand curve.
- Running example: the market for **Tim Hortons medium coffee** in Canada.

Figure 3.1 – Demand Schedule and Demand Curve

Tim Hortons Medium Coffee Demand Schedule

Price ($/cup)	Qty Demanded (M cups/week)
$3.00	10
$3.50	9
$4.00	8
$4.50	7
$5.00	6

Source: Illustrative; based on Statistics Canada CPI data



Demand Schedule and Demand Curve: Key Concepts

- **Demand schedule:** a table showing the relationship between the **price** and the **quantity demanded**.
- **Demand curve:** a curve showing the same relationship graphically.
- When drawing the demand curve, we hold all other factors constant — **ceteris paribus**.
- **Ceteris paribus:** "all else equal" — when analyzing price and quantity demanded, other variables are held constant.

The Law of Demand

- **Quantity demanded:** the amount of a good a consumer is willing and able to purchase at a given price.
- **Law of demand:**
  - Holding everything else constant, when the price of a product falls, the quantity demanded increases.
  - When the price rises, the quantity demanded decreases.
- **Canadian data:** When Tim Hortons raised medium coffee prices by ≈10% in 2022–2023, same-store transaction counts fell in the short run (Restaurant Brands International earnings, 2023).

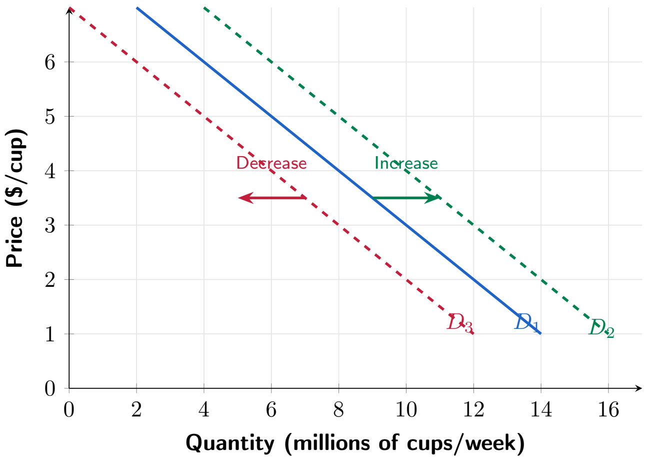
What Explains the Law of Demand?

- When the price of a good falls, two key effects occur:
  - **Substitution effect:** consumers substitute toward the good whose price has fallen.
  - **Income effect:** consumers have more real purchasing power, which raises quantity demanded.
- **Substitution effect:** the change in quantity demanded resulting from a change in price relative to other goods, *holding purchasing power constant*.
- **Income effect:** the change in quantity demanded resulting from a price change affecting consumers' *purchasing power*.

Canadian Example: Substitution and Income Effects

- Suppose Tim Hortons medium coffee falls from \$4.00 to \$3.00.
- **Substitution effect:**
  - Coffee is now cheaper relative to Starbucks lattes, Red Bull, or tea — consumers substitute toward Tim Hortons coffee.
- **Income effect:**
  - A daily coffee drinker saves ≈\$365/year — that extra real income allows more cups per week.

Figure 3.2 – Shifting the Demand Curve



Shifts in Demand: Key Distinction

- A change in something **other than price** causes the **entire demand curve to shift**.
- $D_1 \to D_2$ (right): an increase in demand — more coffee demanded at every price.
- $D_1 \to D_3$ (left): a decrease in demand — less coffee demanded at every price.
- The quantity demanded changes at **every possible price** when the curve shifts.

Variables That Shift Market Demand

- **Income** of consumers
- **Prices of related goods** (substitutes and complements)
- **Tastes**
- **Population and demographics**
- **Expectations** about future prices

Changes in Income: Normal vs. Inferior Goods

- **Normal good:** demand increases as income rises (e.g., restaurant meals, new cars, Starbucks lattes).
- **Inferior good:** demand decreases as income rises (e.g., instant noodles, transit bus passes).
- **Canadian context:**
  - Tim Hortons brewed coffee may behave as a **normal good** overall, but an **inferior good** relative to specialty cafes — as incomes rose in 2010–2019, Starbucks expanded in Canada while Tims' transaction growth slowed (Restaurants Canada, 2019).

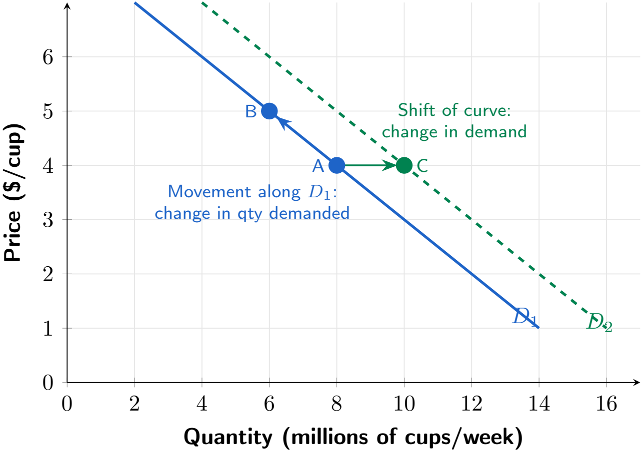
Changes in the Price of Related Goods

- **Substitutes:** goods that can be used for the same purpose.
    - Examples: Tim Hortons coffee and Starbucks latte; Ford F-150 and RAM 1500; WestJet and Air Canada.
- **Complements:** goods used together.
    - Examples: coffee and cream; hot dogs and buns; smartphones and data plans.
- If the price of a Starbucks latte rises, demand for Tim Hortons coffee **increases**.
- If the price of coffee cream rises, demand for Tim Hortons coffee **decreases**.

Changes in Tastes, Population, and Expectations

- **Tastes:** growing health concerns reduce demand for sugary drinks; specialty coffee trends shift demand toward dark roast and espresso.
- **Population and demographics:** Canada's population grew from 38M to 41M between 2021 and 2024 (Statistics Canada, 2024) — more Canadians means more coffee drinkers.
- **Expectations:** if consumers expect coffee prices to jump next month (e.g., due to a drought in Brazil), they increase demand **today**.

Figure 3.3 – Change in Demand vs. Change in Quantity Demanded



Change in Demand vs. Change in Quantity Demanded

- A change in the **price of the product** causes a **movement along** the demand curve — this is a change in quantity demanded.
- Any other change affecting demand causes the **entire demand curve to shift** — this is a change in demand.
- **Common mistake:** saying "the decrease in price causes demand to increase." The decrease in price causes a movement along the curve, not a shift.

3.2 Learning Objective

Goal

Discuss the variables that influence supply.

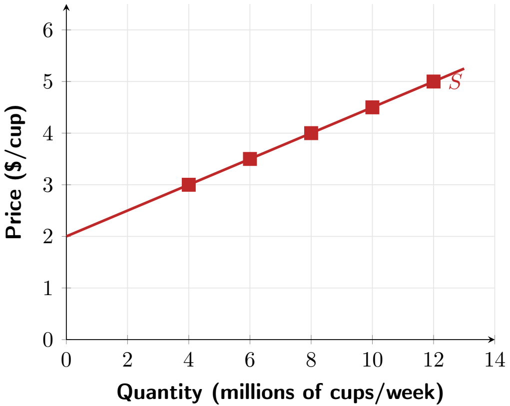
- Now we examine decisions of **firms**: how much of a product to provide at various prices.
- **Market supply:** the supply from all producers of a given good or service.
- We continue with the Tim Hortons medium coffee market.

Figure 3.4 – Supply Schedule and Supply Curve

Tim Hortons Medium Coffee Supply Schedule

Price ($/cup)	Qty Supplied (M cups/week)
$3.00	4
$3.50	6
$4.00	8
$4.50	10
$5.00	12

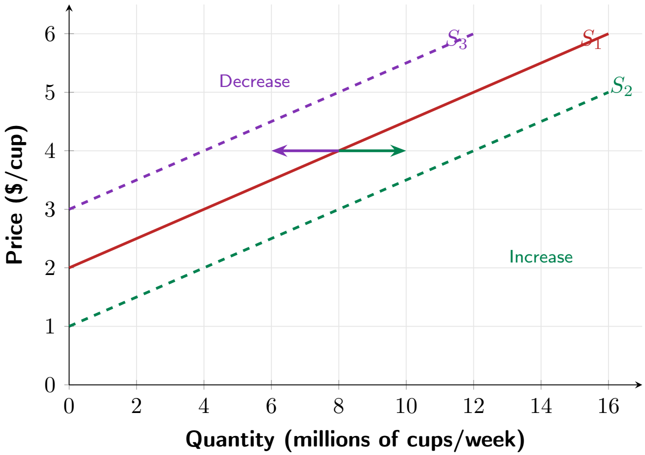
Source: Illustrative; based on Statistics Canada data



The Law of Supply

- **Quantity supplied:** the amount of a good a firm is willing and able to supply at a given price.
- **Law of supply:**
  - Holding everything else constant, increases in price cause **increases** in quantity supplied.
  - Decreases in price cause **decreases** in quantity supplied.
- **Why?** Higher prices make production more profitable, attracting more supply.
- **Canadian context:** When canola prices surged in 2021–2022, Prairie farmers allocated more acres to canola production (Statistics Canada, Field Crop Reporting Series).

Figure 3.5 – Shifting the Supply Curve



Variables That Shift Market Supply

- **Prices of inputs** (labour, coffee beans, milk, paper cups)
- **Technological change** (brewing equipment efficiency)
- **Prices of substitutes in production**
- **Number of firms** in the market
- **Expected future prices**

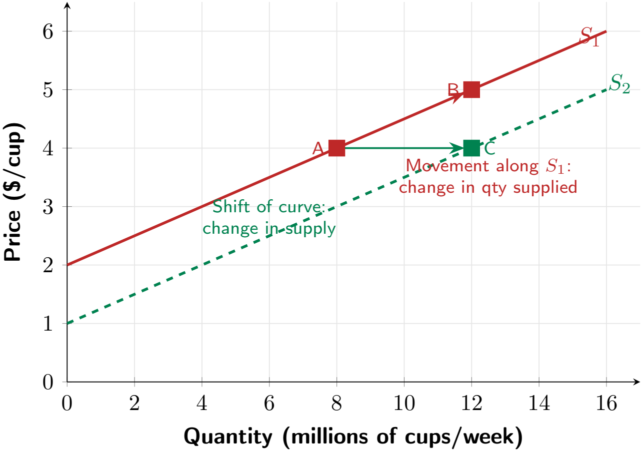
Changes in Prices of Inputs

- **Inputs:** anything used in production — for coffee: arabica beans, milk, labour, paper cups, electricity.
- An increase in input prices decreases profitability $\Rightarrow$ **decrease in supply** (curve shifts left).
- **Canadian example:** Global arabica coffee bean prices rose $\approx$70% in 2024–2025 due to droughts in Brazil, raising costs for all Canadian coffee chains and reducing supply (ICO Coffee Report, 2025).

Technological Change and Other Supply Shifters

- **Technological change:** new, more efficient brewing systems $\Rightarrow$ **increase in supply**.
- **Substitutes in production:** an Ontario farmer can plant corn or soybeans.
  - If soybean prices rise, farmers shift to soybeans $\Rightarrow$ **decrease in corn supply**.
- **Complements in production:** beef and leather (more cattle farming raises supply of both).
- **Number of firms:** more coffee shops entering the market $\Rightarrow$ **increase in supply**.
- **Expected future prices:** if coffee prices expected to rise, firms may withhold supply now.

Figure 3.6 – Change in Supply vs. Change in Quantity Supplied



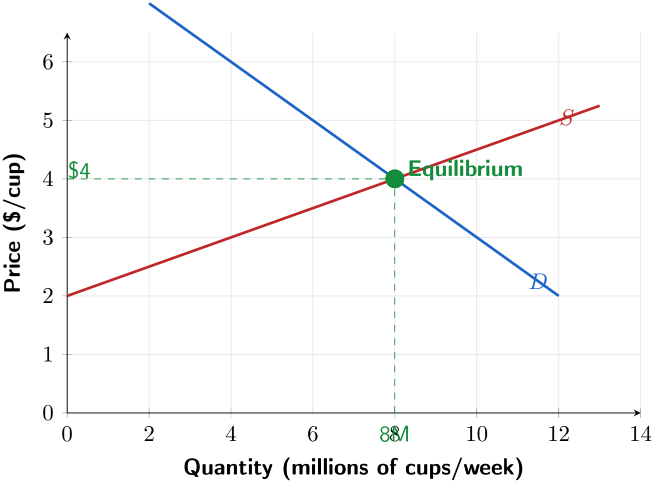
3.3 Learning Objective

Goal

Use a graph to illustrate market equilibrium.

- **Market equilibrium:** quantity demanded equals quantity supplied.
- In perfectly competitive markets, this is called a **competitive market equilibrium**.
- The equilibrium price is the price that *clears* the market.

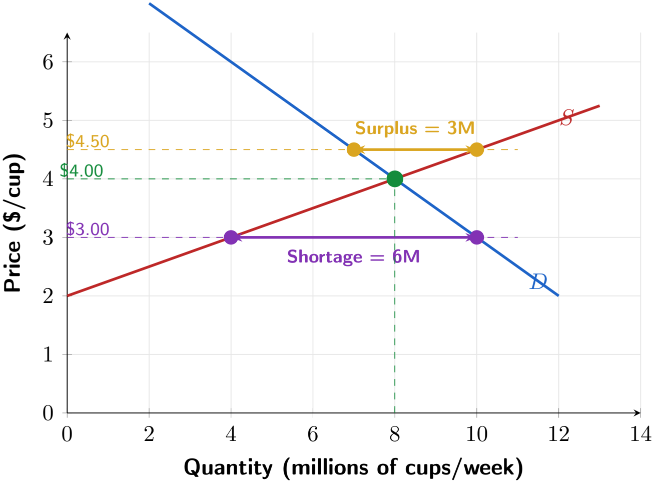
Figure 3.7 – Market Equilibrium: Tim Hortons Coffee



Reading the Equilibrium

- At a price of **$4.00/cup**:
  - Consumers want to buy **8 million** cups/week.
  - Producers want to sell **8 million** cups/week.
- The **equilibrium price** is $4.00/cup; the **equilibrium quantity** is 8 million cups/week.
- At any other price, the market is out of balance — either a surplus or a shortage exists.

Figure 3.8 – Surplus and Shortage



Surplus and Shortage

- **Surplus** (P = \$4.50): $Q_s = 10$M $> Q_d = 7$M — surplus of 3M cups/week.
    - Sellers compete among themselves $\Rightarrow$ price is bid **down** toward \$4.00.
- **Shortage** (P = \$3.00): $Q_d = 10$M $> Q_s = 4$M — shortage of 6M cups/week.
    - Buyers compete among themselves $\Rightarrow$ price is bid **up** toward \$4.00.
- Price is determined by the **interaction** of buyers and sellers — neither group alone controls it.

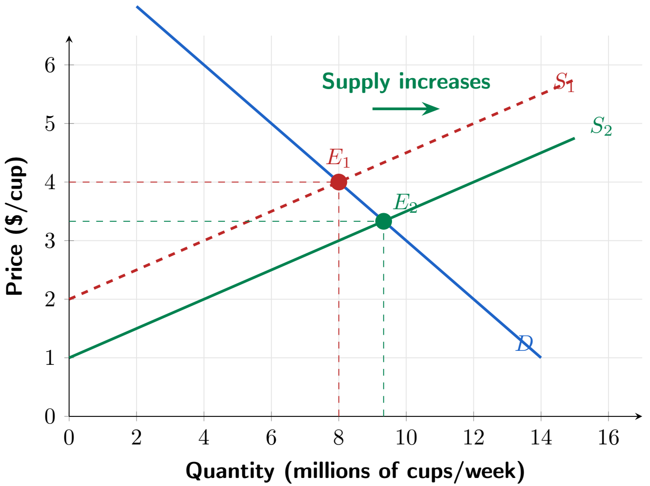
3.4 Learning Objective

Goal

Use demand and supply graphs to predict changes in prices and quantities.

- We typically observe the current price and quantity, but do *not* observe the whole curves.
- The model predicts **directional** changes in price and quantity when curves shift.

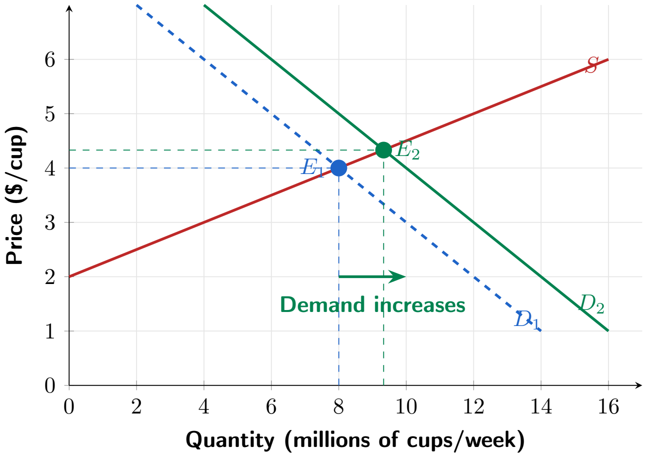
Figure 3.9 – Effect of an Increase in Supply



Effect of an Increase in Supply

- New coffee chains enter the Canadian market (e.g., Dutch Bros. expanding into Canada).
- Supply shifts **right** from S_1 to S_2.
- **Result:**
  - Equilibrium price **falls** (\$4.00 $\rightarrow$ \$3.33).
  - Equilibrium quantity **rises** (8M $\rightarrow$ 9.3M cups/week).
- We only predict *directions* — the exact magnitude requires knowing the full curves.

Figure 3.10 – Effect of an Increase in Demand



Effect of an Increase in Demand

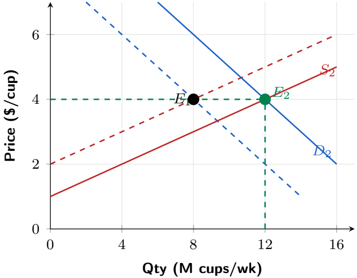
- Canada's population grew by ≈3 million in 2022–2024 (Statistics Canada) — more coffee drinkers.
- Demand shifts **right** from D_1 to D_2.
- **Result:**
  - Equilibrium price **rises** (\$4.00 $\rightarrow$ \$4.33).
  - Equilibrium quantity **rises** (8M $\rightarrow$ 9.3M cups/week).
- Both price and quantity increase when demand increases and supply is unchanged.

Table 3.3 – How Shifts Affect Equilibrium P and Q

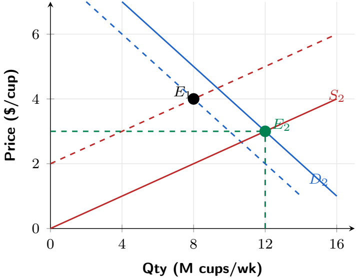
Change	Eq. Price P	Eq. Quantity Q
Demand $\uparrow$, supply unchanged	Rises	Rises
Demand $\downarrow$, supply unchanged	Falls	Falls
Supply $\uparrow$, demand unchanged	Falls	Rises
Supply $\downarrow$, demand unchanged	Rises	Falls
Demand $\uparrow$ and supply $\downarrow$	Rises	?
Demand $\downarrow$ and supply $\uparrow$	Falls	?
Demand $\uparrow$ and supply $\uparrow$	?	Rises
Demand $\downarrow$ and supply $\downarrow$	?	Falls

Figure 3.11 – Both Demand and Supply Shift

Case 1: Demand shifts more



Case 2: Supply shifts more



Both cases: quantity **rises**. Price: **ambiguous** — depends on which shift is larger.

Shifts of Curves vs. Movements Along Curves

- Suppose an **increase in supply** occurs.
  - Equilibrium quantity increases; equilibrium price decreases.
- It is tempting to say "the decrease in price causes an increase in demand" — **incorrect**.
- The decrease in price causes a **movement along the demand curve**, not a shift.
- The demand curve already describes how much consumers want to buy at **any given price**.
- **Canadian example:** When new discount airlines entered the YVR–YYZ route, fares fell and more passengers flew — that is a movement along demand, not a demand shift.

Things to Remember

- A **change in price** causes a movement **along** the demand or supply curve.
- Any **other change** shifts the entire curve.
- In equilibrium, $Q_d = Q_s$; surpluses push price **down**; shortages push price **up**.
- When both curves shift, one effect is **ambiguous** — need to know relative magnitudes.
- Markets coordinate millions of buyers and sellers through prices — no central planner required.

Chapter Summary

- **Demand** follows the law of demand; explained by substitution and income effects. Shifts when income, prices of related goods, tastes, population, or expectations change.
- **Supply** follows the law of supply. Shifts when input prices, technology, production substitutes/complements, number of firms, or expected future prices change.
- **Market equilibrium** occurs where $Q_d = Q_s$. Surpluses drive price down; shortages drive it up.
- The demand and supply model predicts **directional** changes in equilibrium P and Q when curves shift.
- **Data:** Statistics Canada CPI, Bank of Canada, Restaurant Brands International earnings.

Thank you!

Questions?

Data sources: Statistics Canada, Bank of Canada, NRCan, Restaurant Brands International